

Machine Learning Project: To bee or not to bee

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Contributions

- Lianghong LI: project lead; implemented feature extraction, model training, validation, final prediction pipeline, and report integration.
- Zekai YAN: support contribution for data checking, visualization review, and report proofreading.
- Muzi LI: support contribution for reproducibility checking, documentation review, and result verification.

Dataset and Data Issue

Training labels are normalized to ID, bug type, and species. The supervised target is bug type; species is used only for visualization and descriptive comments. ID 154 was excluded because train/masks/binary_154.tif is unavailable. No artificial mask was created, estimated, manually drawn, or imputed. The effective training set is 249 samples.

Bug type counts after exclusion: {'Bee': 115, 'Bumblebee': 100, 'Butterfly': 15, 'Hover fly': 9, 'Wasp': 9, 'Dragonfly': 1}. Largest species group: *Bombus hortorum*.

Feature Extraction

All features are computed from the image and segmentation mask. Pixels outside the mask are excluded from color and texture statistics so that the background does not dominate the representation.

- Color symmetry compares the left-right RGB difference inside the bug bounding box where both original and flipped masks are foreground.
- Shape symmetry flips the binary mask crop left-right and measures normalized foreground overlap.
- Bug pixel ratio is $\text{bug_pixels} / \text{image_pixels}$.
- RGB features are min, max, mean, median, and standard deviation for R, G, and B inside the mask.
- Morphology features describe area, perimeter, circularity, aspect ratio, extent, solidity, orientation, eccentricity, and axis lengths.
- Hu moments provide global shape descriptors that are relatively stable under translation, scale, and rotation.
- HSV/LAB features complement RGB by separating perceptual color properties and lightness.
- Grayscale contrast and edge density capture texture and local structure that can help separate visually similar bug types.

Supervised Models

The target is bug type, not species. Handcrafted features are used for the main task because the assignment requires a mask-based feature pipeline. Macro-F1 is the primary metric because class imbalance makes accuracy alone insufficient. Frequent-class stratified CV is used because singleton classes make full-label stratified CV invalid; full-data metrics are descriptive only.

model	accuracy	macro_f1	weighted_f1	status
mlp	0.871	0.755	0.868	ok
svm_rbf	0.863	0.754	0.859	ok
logistic_regression	0.851	0.698	0.856	ok
extra_trees	0.847	0.618	0.827	ok
gradient_boosting	0.798	0.597	0.790	ok
knn	0.786	0.583	0.766	ok
random_forest	0.819	0.564	0.796	ok
dummy_most_frequent	0.464	0.127	0.294	ok

Numerically highest CV macro-F1 model: mlp (0.7553521749357212). Final recommended submission model: svm_rbf (0.7535392750651922). SVM RBF selected by robustness tie-breaker because its macro-F1 is within 0.005 of MLP and it is simpler/stabler for small imbalanced data.

- DummyClassifier is the imbalance baseline. Macro-F1: 0.126722.
- LogisticRegression is the linear non-ensemble baseline. Macro-F1: 0.697796.
- SVM RBF models smooth nonlinear boundaries and is the final robust recommendation. Macro-F1: 0.753539.
- KNN tests local neighborhood agreement. Macro-F1: 0.582899.
- RandomForest, ExtraTrees, and GradientBoosting provide ensemble comparisons. Their macro-F1 values are 0.563813, 0.618401, and 0.597056.
- MLP is a shallow neural network on extracted features only, not raw images. Macro-F1: 0.755352.

Clustering

method	cluster_count	silhouette	adjusted_rand_index	normalized_mutual_information
kmeans	6.000	0.113	0.149	0.235
agglomerative	6.000	0.077	0.076	0.181
dbscan	0.000		0.000	0.000

KMeans gives the strongest but still moderate/low clustering scores. AgglomerativeClustering finds related but imperfect hierarchical groups. DBSCAN marks all samples as noise under the selected settings. These results indicate useful feature structure but not perfectly separated natural clusters.

Final Prediction Pipeline

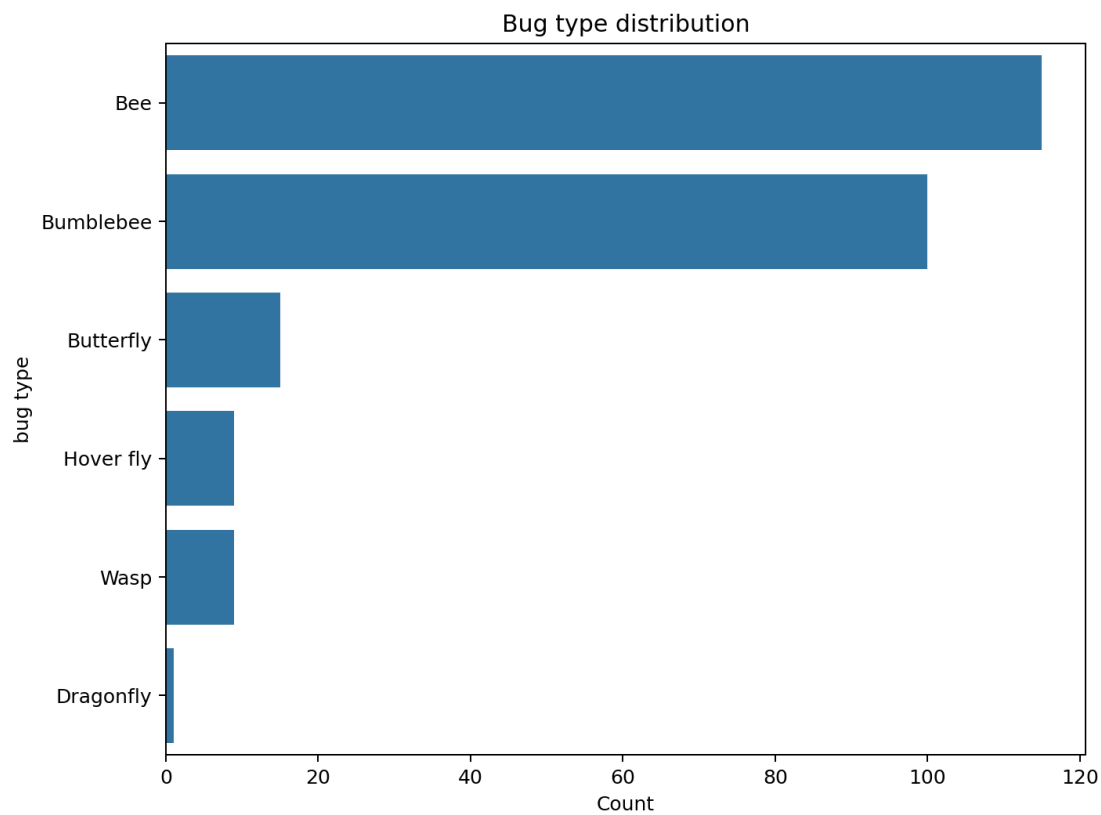
The official test images and masks for IDs 251-347 were released and processed. The final svm_rbf model was applied to 97 test samples. results/submission.csv was generated with exactly two columns: ID and bug type.

The CSV was validated for 97 rows, IDs 251-347, no duplicate IDs, no missing predictions, and valid bug type labels. Prediction distribution: {'Bee': 46, 'Bumblebee': 38, 'Butterfly': 6, 'Wasp': 5, 'Hover fly': 2}. No test accuracy is reported because test labels are unavailable.

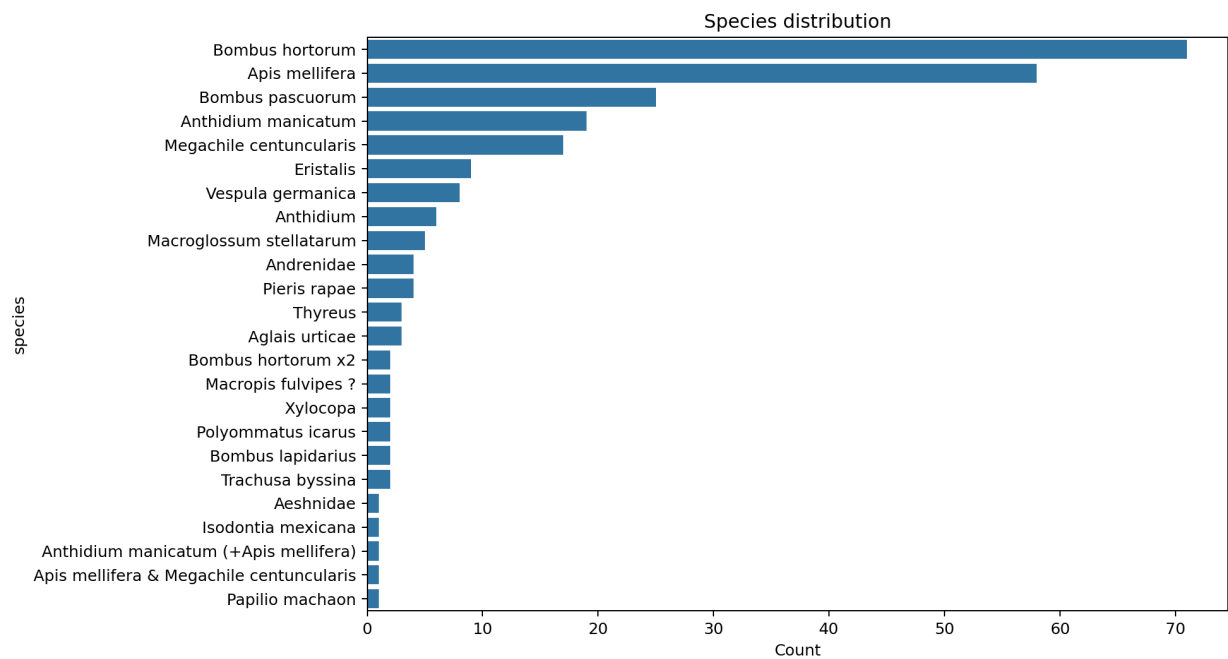
Generated Figures

The bug type distribution highlights imbalance, with Bee as the largest class. Species plots show finer-grained label imbalance but species is not the target. PCA, t-SNE, and isomap projections show partial grouping with overlap, supporting the need for supervised classification.

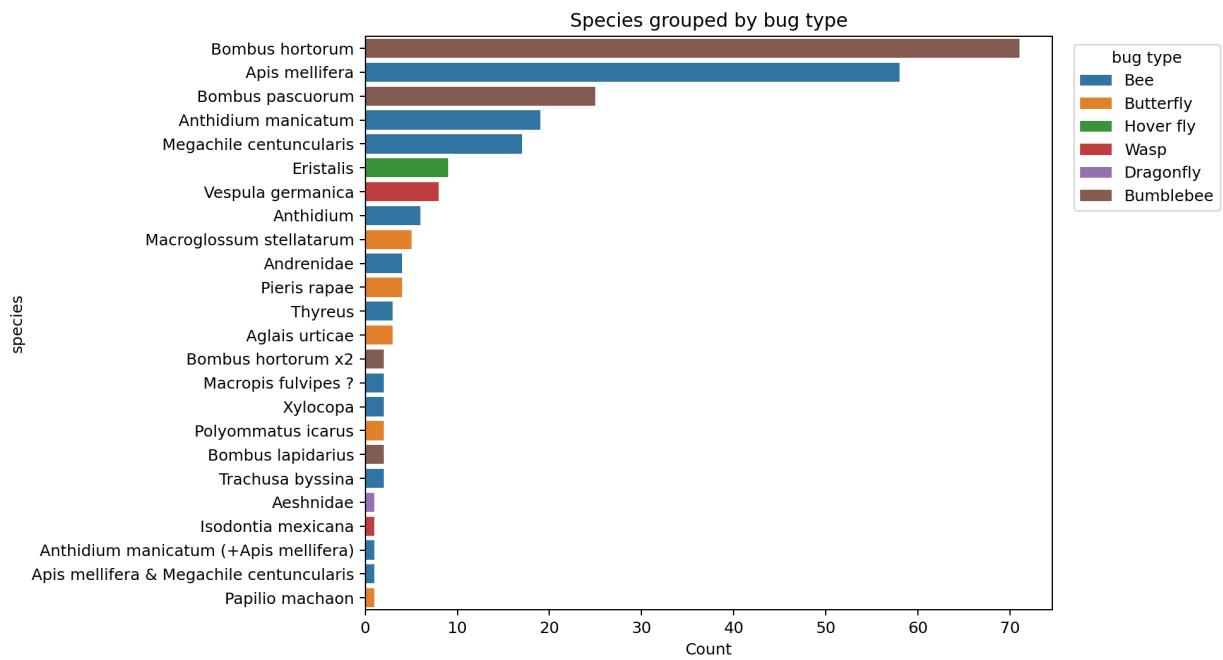
Bug type distribution



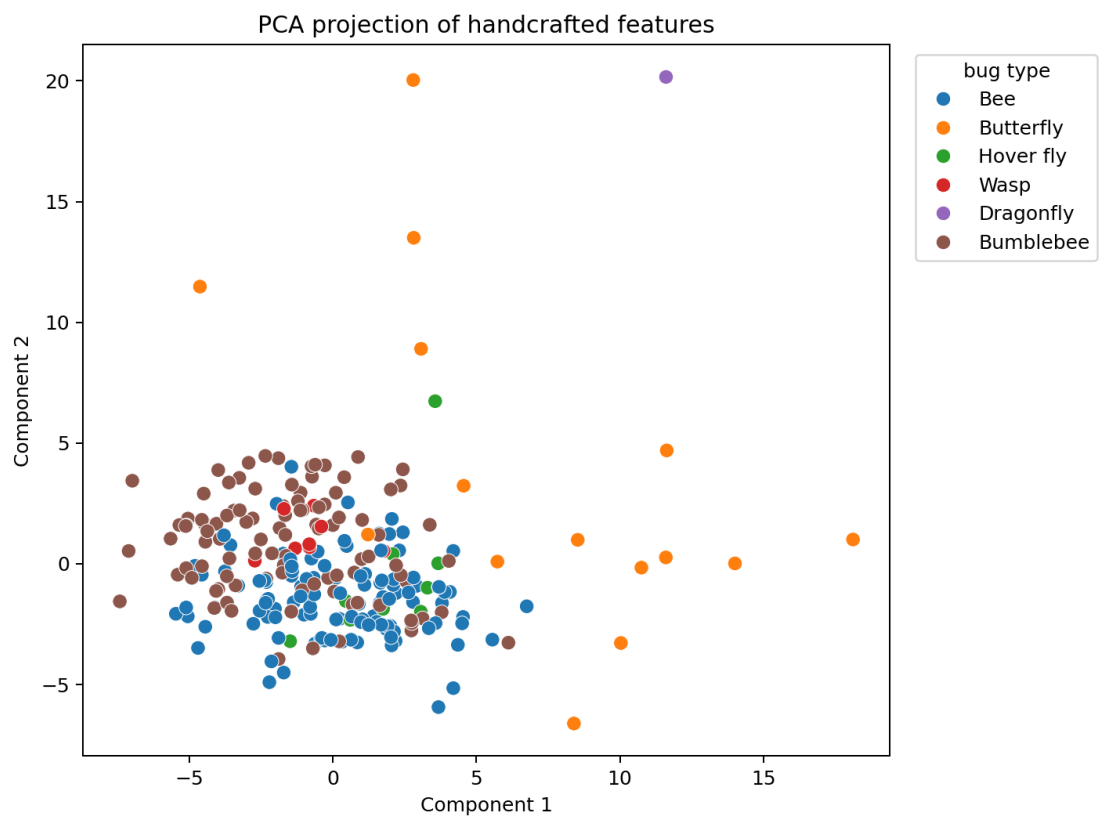
Species distribution



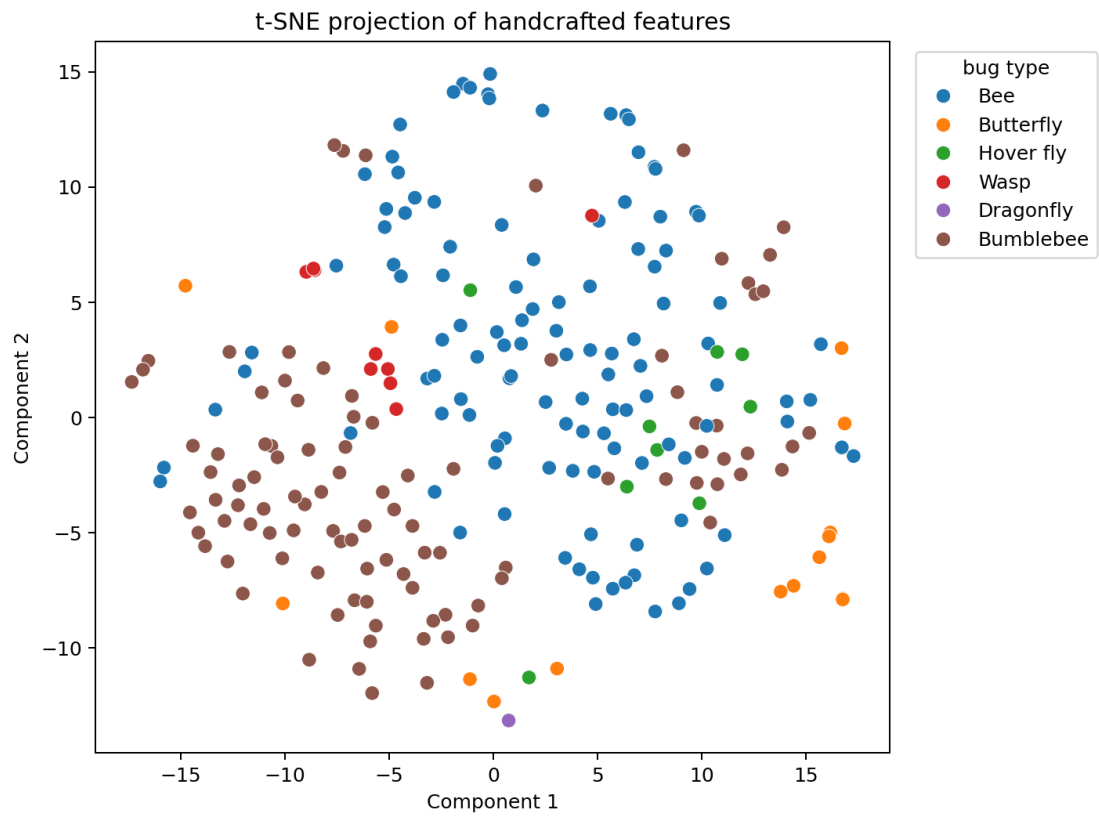
Species grouped by bug type



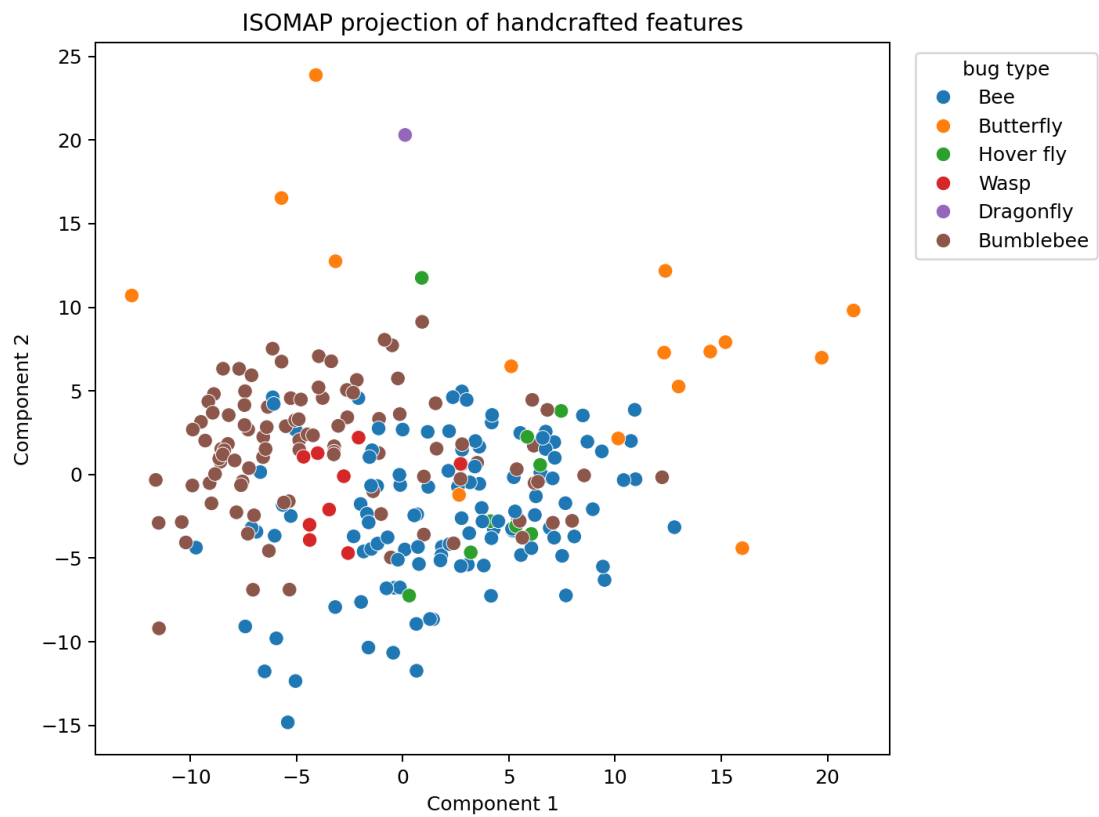
PCA projection



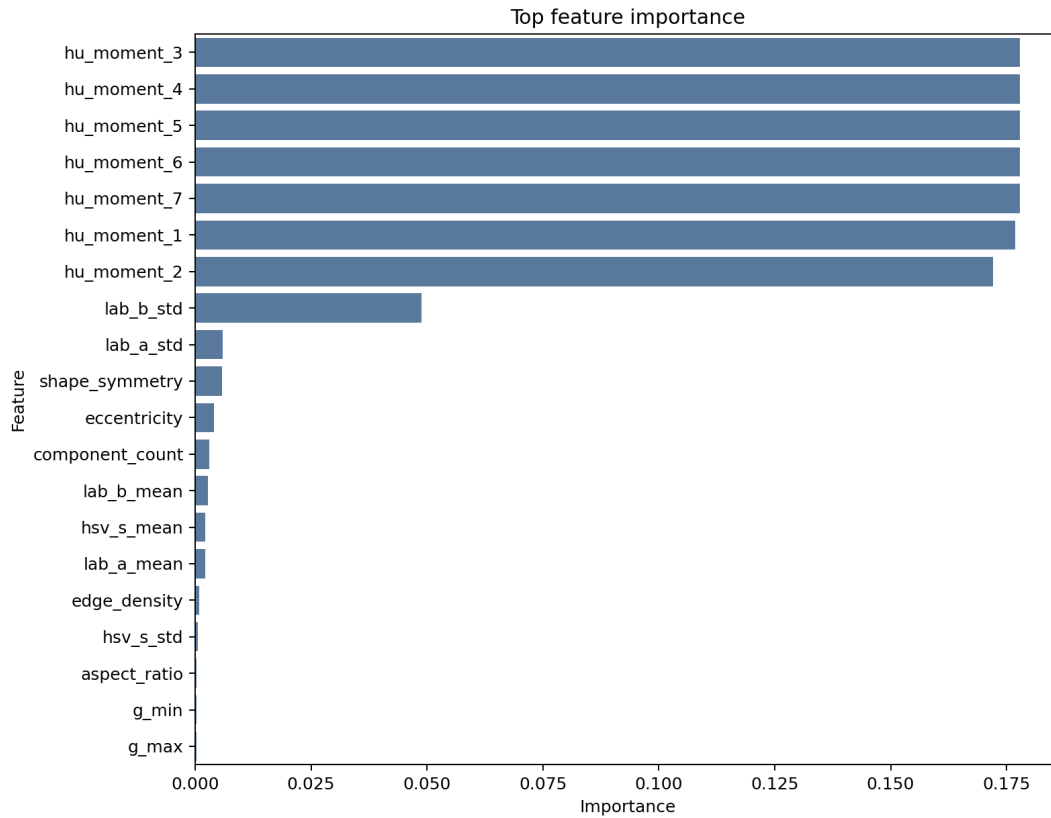
t-SNE projection



Nonlinear projection



Feature importance



Reproducibility

Run: `python3 -m src.data_loader --validate --split train; python3 -m src.build_features --split train; python3 -m src.train_models; python3 -m src.clustering; python3 -m src.visualize; python3 -m src.make_report; python3 -m src.data_loader --validate --split test; python3 -m src.build_features --split test; python3 -m src.predict; python3 -m src.validate_submission results/submission.csv`. Test data is not used for model selection or hyperparameter tuning.